ECA5403-ELECTROMAGNETIC FILEDS AND WAVES

Hemavathi .M

192512222

C02-7

EXPERIMENT 7

AIM:
To determine and plot the capacitance of an isolated sphere and a concentric spherical capacitor for different radii and dielectric constants using MATLAB.

Test case (i) Capacitance of an Isolated Sphere

Program:

```
clc;

clear;

epsilon0 = 8.854e-12;

a = input('Enter radius of isolated sphere (m): ');

er = input('Enter relative permittivity: ');

 epsilon = epsilon0*er;

C = 4*pi*epsilon*a;

 disp('Capacitance of isolated sphere (F) =');

disp(C);
```

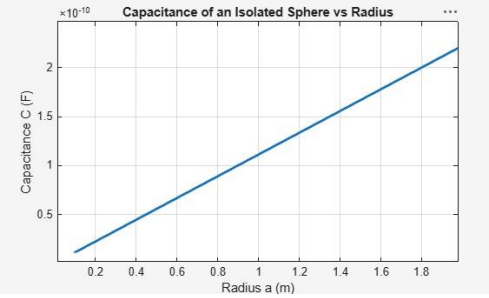
OUTPUT:

```
Enter radius of isolated sphere (m): 0.50
Enter relative permittivity: 1
Capacitance of isolated sphere (F) =
   5.5631e-11
```

Test case (ii) Variation of Isolated-Sphere Capacitance with Radius.
program:

```
clc;
clear;
epsilon0 = 8.854e-12;
a = input('Enter radius of isolated sphere (m): ');
er = input('Enter relative permittivity: ');
epsilon = epsilon0*er;
C = 4*pi*epsilon*a;
disp('Capacitance of isolated sphere (F) =');
disp(C);
```

OUTPUT:



Test case (iii) Capacitance of a Concentric Spherical Capacitor**:**
Program:

```
clc;
 clear;
epsilon0 = 8.854e-12;
 a = input('Enter inner radius a (m): ');
 b = input('Enter outer radius b (m): ');
er = input('Enter relative permittivity: ');
epsilon = epsilon0*er;
C = 4*pi*epsilon*a*b/(b-a);
disp('Capacitance of concentric spherical capacitor (F) =');
disp(C);
```

OUTPUT:

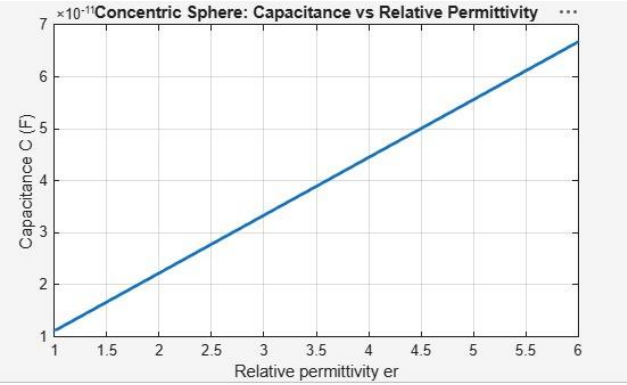
```
Enter inner radius a (m):  0.05
Enter outer radius b (m):  0.10
Enter relative permittivity: 2.5
Capacitance of concentric spherical capacitor (F) =
   2.7816e-11

>>
```

Test case (iv) Effect of Outer Radius and Relative Permittivity:
Program:

```
clc;
clear;
 epsilon0 = 8.854e-12;
 a = 0.05;
er = 2.5;
b = linspace(0.06,0.20,100);
C1 = 4*pi*epsilon0*er.*a.*b./(b-a);
figure;
 plot(b,C1,'LineWidth',2);
 grid on;
xlabel('Outer radius b (m)');
ylabel('Capacitance C (F)');
```

```
title('Concentric Sphere: Capacitance vs Outer Radius');
er = linspace(1,6,100);
b = 0.10; C2 = 4*pi*epsilon0.*er.*a.*b./(b-a);
figure;
plot(er,C2,'LineWidth',2);
grid on;
xlabel('Relative permittivity er');
ylabel('Capacitance C (F)');
title('Concentric Sphere: Capacitance vs Relative Permittivity');
```



Result :

Thus the program was verified for Capacitance of an Isolated and Concentric Sphere with Different Inner and Outer Radius and Dielectric Constant using MATLAB successfully.

Objective	**Assessment Criterion**	**Excellent – 4**	**Good – 3**	**Satisfactory – 2**	**Needs Improvement – 1**	**Weighting**	**Marks given**	**Marks obtained**
Define spherical geometry and dielectric	Define spherical geometry and dielectric	Correctly identifies isolatedsphere radius or inner/outer radii and medium permittivity	Geometry is correct with minor omission.	Some dimensions or material properties are unclear	Uses invalid geometry or material inputs.	25	2.5	
Implement spherical capacitance equations	Implement spherical capacitance equations	Applies the appropriate isolated/concentric sphere formulas correctly.	Formulas are mostly correct with minor errors.	Partial calculations require guidance.	Uses incorrect equations.	25	2.5	

Analyze radius and permittivity effects	Analyze radius and permittivity effects	Compares cases and validates the expected capacitance trends against theory.	Analysis is mostly correct with minor omissions.	Comparisons or validation are incomplete.	No valid trend analysis is shown.	30	3	
Present results and conclusions	Present results and conclusions	Reports geometry, units, calculations and conclusions clearly.	Report is clear with minor gaps	Report lacks some key information.	Results are unclear or missing.	20	2	
TOTAL						100	10	